

(1-12, 17-19) Basil. v. Lin. Wan Ex. 6.2

Q1)

(a).

Ans: $\|u\| = (1, -3) \cdot (1, -3)$
 $= 1 + 9 \rightarrow 10$

$|u| = (2, 4) \cdot (2, 4) = 20$

$|u+v| = \sqrt{10} \cos \theta = \sqrt{10}$
 $= (2, 4) \cdot (2, 4) = 20$

$\|u, v\| = 2 - 12$

(b). $u = (-1, 5, 2), v = (2, 4, -9)$

$|u+v| = -2 + 20 = 18$

$= 0$ orthogonal

(c). $|u+v| = -3 - 3 \rightarrow -6$

$\|u\| = \sqrt{2} \cos \theta = -6$

$\|u\| = \sqrt{9+9+9} \rightarrow 6 \sqrt{3}$

$\theta =$

Q2) done

Q3). $p = -1 + 5x + 2x^2, q = 2 + 4x - 9x^2$

$\|p\| = (2) + 25 + 4 \rightarrow \sqrt{30}$

$\|q\| = (4) + 16 + 81 \rightarrow \sqrt{101}$ orthogonal

$\|p+q\| = \sqrt{-2+20-18} \rightarrow 0$

Q4) done

Q6) done

Q5). $\|A\| = \begin{bmatrix} 2 & 4 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 2 & 6 \\ 1 & -3 \end{bmatrix}$

$= 4 + 36 + 1 + 9 = 50$

$\|B\| = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix} = 9 + 4 = 13$

$\|A+B\| = 4 + 12 + 1 = 17$

Q7) $(2) = -4 + 6 - 2$

$= 0$ orthogonal

(1) $-2 - 2 - 2$

not orth

(c) orthogonal

(d) done

Q9). $0 - 2 + 2$

orthogonal

Q10). $3 - 6 + 2 = 0$ orthogonal

Q11). $-6 + 6 \rightarrow 0$ orthogonal

Q12). $5 - 3 - 2 \rightarrow 0$ orthogonal

Q13). $p_1 = 2 + 6x + 6x^2$

$p_2 = 2 + 5x + 3x^2$

$p_3 = 1 + 2x + 3x^2$

so $\rightarrow (p_1, p_2) = 0$

$(p_2, p_1) = 0$

$(p_3, p_3) = 0$

$22 + 56 + 18 = 0$

$32 + 20 + 9 = 0$

$\lambda = -19$ const. not exist.

$u = 4$

Q14). $Au \rightarrow Av$

$= \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix}$

$= \begin{bmatrix} 9 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -3 \end{bmatrix} \rightarrow 18 - 18 = 0$

GENIUS

(1-

Q1)

Q6

Q1)

Q3).

=

(b).

=

Q1)

(a). $\int$

Q5). Q6

Q7). $(v_1$

$v_2,$

$(u_1,$

$u = (1, -$

Q1)

Q1)

orthogonal